Keywords of Plotfunction

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1.0.4 brown

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Dark blue.

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Dark green

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light blue

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light green

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light grey

1.0.14 magenta

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1.0.17 pink

1.0.18 purple

1.0.19 red

1.0.20 skyblue

1.0.21 violet

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1.0.23 yellow

2 Commands

2.1 analyze:

Usage: **analyze [#nr = 1] [x_1 x_2] [prec = 1e-9] [iterations = number of pixels of width][decimals = 4]**
Analyses the function number nr in the [x_1 , x_2] interval. prec is the precision used in the algorithms. Iteration is the number of evaluations of that function to count the changes of sign in the given interval. decimals are the number of decimals for rounding all results.

2.2 arclength:

Usage: **arclength [#nr = 1 / "function"] [decimals=4 iterations=50 precision=1e-8] [x_1 x_2]**
Calculates the arc length of the function #nr or "function" in the interval [x_1 x_2] if given, or the x axis inteval (default). The approximate result is rounded by the given decimals. Iterations and precision are needed in the used algorithms to approximate the integral value (Kepler Simpson).

2.3 calculate:

Usage: **calculate arithmetic expression**

2.4 derive:

Usage **derive [#nr = 1]**
Derives function to x (x_1).

2.5 evaluate:

Usage **evaluate [#nr = 1] expressions [decimals = 4]**
Evaluates the #nr-th. function value of the expressions and rounds the result to the given decimals.

2.6 integral:

Usage: **integral [#nr = 1 / "term"] [decimals iterations precision] [x_1 x_2]**
Numerical approximation (if possible also exact computation via a stamm-function) of the integral of the #nr-th function (or the function by the given term) in the interval [x_1 x_2]. The result is rounded to the given decimals and the algorithms use the given number or iterations and precision (Kepler-Simpson-Algorithm).

2.7 interpolation:

Usage: **interpolation #nr / points for f [; points for f'] [;points for f"]**
Polynomial interpolation of the #nr-th geometric object (points or polygon) or the given points for f and eventually f' and f".

2.8 intersection:

Usage: **intersection [#nr1 = 1] #nr2 [x_1 x_2] [precision] [iterations] [decimals]**
Computes the approximation (if possible also the exact) points of intersection and intersection angle between the #nr1-th and #nr2-th object in the [x_1 x_2] interval. The coordinates are rounded to the given decimals. The Algorithm (Newton) uses the given precision and number of iterations.

2.9 latex-string:

Usage: **latex-string [#nr = 1]**
Translates the #nr-th function term to a latex string. Example: *latex-string x/exp(x)* would compute the string: $\frac{x}{e^x}$

2.10 move:

Usage: **move [#nr = 1] dx dy**
Moves the #nr-th geometric object dx units to the right (left if dx < 0) and dy units to the top (bottom if dy < 0).

2.11 normal:

Usavge: **normal [#nr = 1] x / x y**
Computes the normal to the #nr-th function at x or from the point (x,y).

2.12 osc-circle:

Usage: **osc-circle [#nr = 1] x**
Computes the osculating circle to the #nr-th function (or parametric curve) at x.

2.13 points-in-graph:

Usage: **points-in-graph [#nr = 1] expression1 (x= / y=); expression2... [;ddecimals [= 4]]**
Draws the points in the graph of the nr-th function wrt the given x-, y- or parameter-expressions.
Example: *points-in-graph x=4; y=sqrt(2); d=2* draws the points to the graph of the first function wrt to x = 4 and y = sqrt(2). The number of decimals for these points is 2.

2.14 reflection:

Usage: **reflection [#nr1 = 1] #nr2 / object**
Reflects the object nr1 to the axis/point defined by the object nr2.

2.15 regression:

Usage: **regression #nr [= 1]/points [degree = regression-degree]**
Computes the regression polynomial of the given degree wrt the given points or the #nr-th geometric object.

2.16 reset-colors:

Resets all the colors to the default colors.

2.17 rotate:

Usage: **rotate [#nr = 1] angle (in degrees) x y (center)**
Rotates the #nr-th geometric object / function by an angle of the given degrees wrt the given center (x,y).

2.18 show-function:

Usage: **show-function [#nr = 1] 0/1**
Shows (if value is 1) or hides (if value is 0) the #nr-th object.

2.19 show-grid:

Usage: **show-grid 0/1**
Shows (if value is 1) or hides (if value is 0) the grid.

2.20 show-x-axis:

Usage: **show-x-axis 0/1**
Shows (if value is 1) or hides (if value is 0) the x-axis.

2.21 show-x-scale:

Usage: **show-x-scale 0/1**
Shows (if value is 1) or hides (if value is 0) the x-scale.

2.22 show-y-axis:

Usage: **show-x-axis 0/1**
Shows (if value is 1) or hides (if value is 0) the y-axis.

2.23 show-y-scale:

Usage: **show-x-scale 0/1**
Shows (if value is 1) or hides (if value is 0) the y-scale.

2.24 table:

Usage: **table [#nr = 1 / "function"] [x_1 x_2] [dx = 0.5] [;]**
Evaluates the function in the values from x_1 to x_2 with an increment of dx. If there is a ; at the end, the values are printed in fractions.

2.25 tangent:

Usage: **tangent [#nr = 1] x / x y**
Computes the tangent to the #nr-th function at x or from the point (x,y).

2.26 taylor-polynomial:

Usage: **taylor-polynomial [#nr = 1] deg [x0 = 0]**
Computes the taylor polynomial or deg-th degree of the nr-th function at x0.

2.27 zero-iteration:

Usage: *zero-iteration [#nr = 1 / "function"] [decimals iterations precision] [x_1 x_2] *
Computes the approximation of a root in the [x_1, x_2] interval of the given object.
The result is rounded to the given decimals. The Algorithm (Newton) uses the given precision and number of iterations.

3 Constants

3.0.1 inf:

stands for infinity.

3.0.2 PI:

stands for the irrational number π.

4 Geometric Objects

4.1 bitmap:

Usage: **bitmap_[nr] x y width height**
Draws the stored bitmap nr on to the position x y (axis-coordinates) with the given with and height in pixels.

4.2 circle:

Usage: **circle x y r [α_1 α_2 [slice]] [<color>]**
Draws a circle with middle-point x y, radius r. If α_1, α_2 is given an arc of

circle is drawn, beginning at the angle α_1 and ending at α_2. If slice = 1 a transparent slice is drawn, if slice = 2 a filled slice is drawn.

4.3 fill:

Usage: **fill x y [<color>]**
Floodfill at the point (x,y) with the given color.

4.4 histogram:

Usage: **histogram points [t = 0 , w = 1] [<color>]**
Draws a histogram wrt the given points. t is the value for the transparency of the background color, where 0 is transparent and 1 is opaque. w is the width of the columns wrt the x-axis range.

4.5 line

Usage: **line point$_1$ point$_2$ [h = 0] <[color,] l>**
Draws a line (or arrow if h != 0) from point$_1$ to point$_2$. h is a float number for the size of the arrowhead (h = 0 for no arrowhead). If l >= 1 the line will be dotted.

4.6 points:

Usage: **points x_1 y_1 x_2 y_2 ... x_n y_n [<color>]**
Draws n points with the given coordinates.

4.7 polygon:

Usage: **polygon x_1 y_1 x_2 y_2 ... x_n y_n [<color>]**
Draws polygon with the n vertices $(x_{1,y1}), \ldots , (x_{n,yn})$.

4.8 rectangle:

Usage: **rectangle x_1 y_1 x_2 y_2**
Draws a rectangle with the left bottom corner $(x_{1,y1})$ and the top right corner $(x_{2,y2})$.

4.9 text:

Usage: **text {[ł] string} x y [<color>]**
Displays a string at the coordinates (x,y).
If latex is installed you can use łto generate a latex string.

4.10 triangle:

Usage: **triangle $\mathbf{x_1}$ $\mathbf{y_1}$ $\mathbf{x_2}$ $\mathbf{y_2}$ $\mathbf{x_3}$ $\mathbf{y_3}$ <[color,] l>**
Draws a triangle with vertices $(x_{1,y1})$, $(x_{2,y2})$, $(x_{3,y3})$.
If l >= 1 the line will be dotted.

5 Functions

5.1 abs:

Absolute value of a real or a complex number.

5.2 arccos:

Arcus cosine function.

5.3 arcosh:

Area hyperbolic cosine function.

5.4 arcsin:

Arcus sine function.

5.5 arctan:

Arcus tangent function.

5.6 arsinh:

Area hyperbolic sine function.

5.7 artanh:

Area hyperbolic tangent function.

5.8 cos:

5.9 cosh:

Hyperbolic cosine.

5.10 exp:

The exponential function.

5.11 log:

Logarithmus Naturalis.

5.12 sin:

Sine function.

5.13 sinh:

Hyperbolic cosine.

5.14 sqrt:

The square root function.

5.15 tan:

The trigonometric tangent function.

5.16 tanh:

Hyperbolic tangent function.

6 Settings

6.1 axis-color:

Usage: **def. color / Red Green Blue**
Use with the defined colors or input three values from 0 to 255 for the red, green and blue part.

6.2 axis-fontsize:

Usage **axis-fontsize unsigned int**

6.3 axis-names:

Usage: **axis-names string string**
Example: *axis-names x y*

6.4 axis-range:

Usage: **axis-range $\mathbf{x_1}$ $\mathbf{x_2}$ [: $\mathbf{y_1}$ $\mathbf{y_2}$]**
Example: *axis-range -3 5 : -2 7* : x values range from -3 to 5, y values from -2 to 7 *axis-range 5 : 10* : x values from -5 to 5, y values from -10 to 10.

6.5 axis-scale:

Usage : **axis-scale sx [sy = sx]**
Example: *axis-scale PI/2 1*

6.6 background-color:

Usage: **background-color def. color / Red Green Blue**
Use with the defined colors or input three values from 0 to 255 for the red, green and blue part.

6.7 font-size:

Usage: **font-size unsigned int**
Sets the font size in pixels for input/output text.

6.8 function-color:

Usage: **function-color [#nr = 1] defined color / Red Green Blue**
Draws the #nr-th function/geometric object with the given defined color or the color defined by the three values from 0 to 255 for the red, green and blue part.

6.9 function-settings:

Usage: **function-settings [#nr = 1]**
Opens a settings dialog for the #nr-th. function or geometric object.

6.10 function-size:

Usage: **function-size [#nr = 1] unsigned number**
Sets the line thickness of the #nr-th. function/geometric object to the given number.

6.11 grid-color:

Usage: **grid.-color def. color / Red Green Blue**
Draws the grid lines in the given defined or the color obtained by the three values for the red, green and blue part (from 0 to 255 each).

6.12 grid-scale:

Usage: **grid-scale sx [sy = sx]**
Example: *grid-scale PI/2 1*

6.13 margin:

Usage: **margin unsigned int**
Sets the margin (offset) of the coordinates system in pixels.

6.14 move-increment:

Usage: **move-increment dx [dy=dx] (p for pixels)**
Move increment values for moving objects or the coordinate system with the arrow-keys.

6.15 panel-size:

Usage: **panel-size sx [sy = sx]**

6.16 parameter-values:

Usage: **parameter-values double ...**
Sets the values for the parameter t of the functions.

6.17 point-decimals:

Usage: **point-decimals int**

6.18 point-size:

Usage: **point-size int**
Sets the size in pixel for points (wrt to style 1 and 2)

6.19 regression-degree:

Usage: **regression-degree unsigned int**
Sets the regression degree for the regression computations.

6.20 select:

Usage: **select [#nr=1]**
Selects (highlights) the nr-th function.

6.21 values-number:

Usage **values-number unsigned int**
The number of points that are evaluated for rendering the functions.

7 Alphabetically

7.1 A

7.1.1 abs

7.1.2 analyze:

7.1.3 arccos

7.1.4 arclength

7.1.5 arcosh

7.1.6 arcsin

7.1.7 arctan

7.1.8 arsinh

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7.2.5 blue

7.2.6 brown

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7.3.2 cos

7.3.3 cosh

7.3.4 cyan

7.4 D

7.4.1 dblue

7.4.2 derive

7.4.3 dgreen